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Public Transport: From NeTeX-Files to Mathematical Graphs

Jörn Schönberger^a, Stefan Fach^b

^a*TU Dresden, Faculty of Transport and Traffic Sciences, 01069 Dresden, Germany*

^b*INAVET – Institut für angewandte Verkehrstelematik GmbH, Löbtau Str. 67, 01159 Dresden, Germany*

Abstract

This paper focuses on the automatic extraction of routable mathematical graphs from open-data sources for the representation of real-world public transport networks. We start with the composition of the information required to define a graph for determining passenger journeys. Next, we explain the content of the used open data sources and demonstrate its appropriateness. Third, we apply our developed tool used for the graph derivation. The generated graphs provide a comprehensive representation of real-world multimodal public transportation networks.

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1. Introduction

Public transport networks are complex systems. A careful system design and process are both necessary. Due to their inherent complexity public transport networks are also an often used test field for algorithms. A complete formal network description is a prerequisite in both situations. Mathematical graphs composed of nodes and arcs are the favored representation tool. However, these data sets are often not available or require complex manual data processing and modification. The main purpose of the here reported research is to demonstrate how such graphs can be extracted from publicly available data sets (open data sources).

This paper addresses the automatic generation of comprehensive mathematical graph parameter sets that represent a specific public transport network. The main research idea is to exploit regularly published and updated open data sets following the Network Timetable Exchange data reference model. All required data can be extracted automatically. In doing so, the time to prepare a complete mathematical graph representation shrinks dramatically to several minutes.

* Jörn Schönberger

E-mail address: joern.schoenberger@tu-dresden.de

The here presented concepts and developed software tools contribute to providing comprehensive graph formulations. The accessibility for both scientists as well as practitioners to these graphs is established.

We compile the necessary knowledge for preparing the automatic generation of mathematical graphs based on NeTeX-files (Section 2). We outline the data found in a NeTeX-files (Section 3). We present software that automatically extracts the relevant node and arc data from a collection of NeTeX files and we report initial results for a combined tram-&-bus network (Section 4).

2. Public transport networks, data reference models and mathematical graphs

The GTFS-reference model is today's most commonly used reference model for public transport network descriptions (Zervaas (2014)). Its concept suffers from the distribution of the line describing data over several flat CSV files. This shortcoming limits the flexibility of this data description mode with respect to special local, regional, or national requirements. As an improved and extended alternative, the reference model **TransModel** has been introduced (Ruckhaus et al. (2023)) and was promoted to become the official and obligatory data interface description for the European Union if it comes to public transport data exchange. Provided data is stored in XML-coded files (NeTeX files), which offers a more flexible data storage scheme.

NeTeX (Network Timetable Exchange) is a CEN-standard for exchanging public transport-related network, timetable, and fare information (Buzna et al. (2023)) in files. A paradigm of NeTeX is to collect all information associated with one service line (route) in one XML file. All public transport operators in the participating EU countries are requested to publish NeTeX files of all operated lines to everyone interested and to update them regularly. Typically, this happens via national access points (e.g., websites). An example list is available under the URL <https://napcore.eu/description-naps/national-access-point/>. The composition of a selection of NeTeX files is equivalent to collecting all network data relevant to the description of a typical multi-modal public transport network, including vehicle stop positions, service lines, transfer information, fare information, and timetables.

Since GTFS and NeTeX originate from different application contexts, the underlying assumptions and requirements used to compile a data feed differ. Knowles et al. (2009) provide a comprehensive discussion about similarities and differences of both reference models. In brief, GTFS mainly aims to provide core network and timetable information associated with a compact local or regional service network, needed to inform passengers and passenger information application systems. In contrast, the primary intention of the NeTeX standard is to host a comprehensive data collection serving a broad variety of applications, e.g. also sophisticated fare information systems as well as long-distance services or vehicle displaying. However, both data models host only static or planned information about the network or the services on offer. To include real-time data they use individual model extensions. These are called GTFS-RT (Aemmer et al. (2022)) and SIRI (<https://transmodel-cen.eu/index.php/siri/>).

A mathematical graph defines a formalized ecosystem for planning the mobility of vehicles and passengers. The detailed information needed depends on the planning purpose and impacts the extent of the included network-describing features. Specifying a node set and an arc set (Gross et al. (2018)) is necessary. Additional details for nodes and arcs can be stored. Nodes represent locations. In the context of public transport networks, these are the positions where vehicles stop to let passengers board or get off (quays). An arc originating from one node and terminating in another represents the opportunity to travel nonstop using a public transport vehicle serving a given and known route. A transfer is an essential ingredient to the origin-to-destination travel path of a passenger. During a transfer, a passenger walks from one quay to another quay at a transfer location (stopplace) to exploit the offer of different service routes. Therefore, it is also crucial to consider transfer options in a mathematical graph description of a public transport network. Due to this requirement, we must add transfer locations and arcs to the graph definition.

Transfers introduce hierarchy among locations (nodes) since quays are subordinate entities of stopplaces. The induced hierarchy requires a specific common numbering scheme. Here, we use IFOPT-numbers (CEN (2012)) as unique identifiers for nodes representing stopplaces and nodes which are quays. An IFOPT-number has the form $A : B : C : D : E$. An example number following this scheme is $de : 04011 : 13925 :: A$. The first component (de) specifies the country in which the node is situated (expressed by the official two-letter country code of the EU: de for Germany). The second component (04011) refers to the administrative region defined by the given NUTS-3-code (here: city of Bremen, Germany). The third component is the identifier of the stopplace assigned by the regional administration (13925 defines the collection of all quays at the transfer location Hauptbahnhof in Bremen). These first

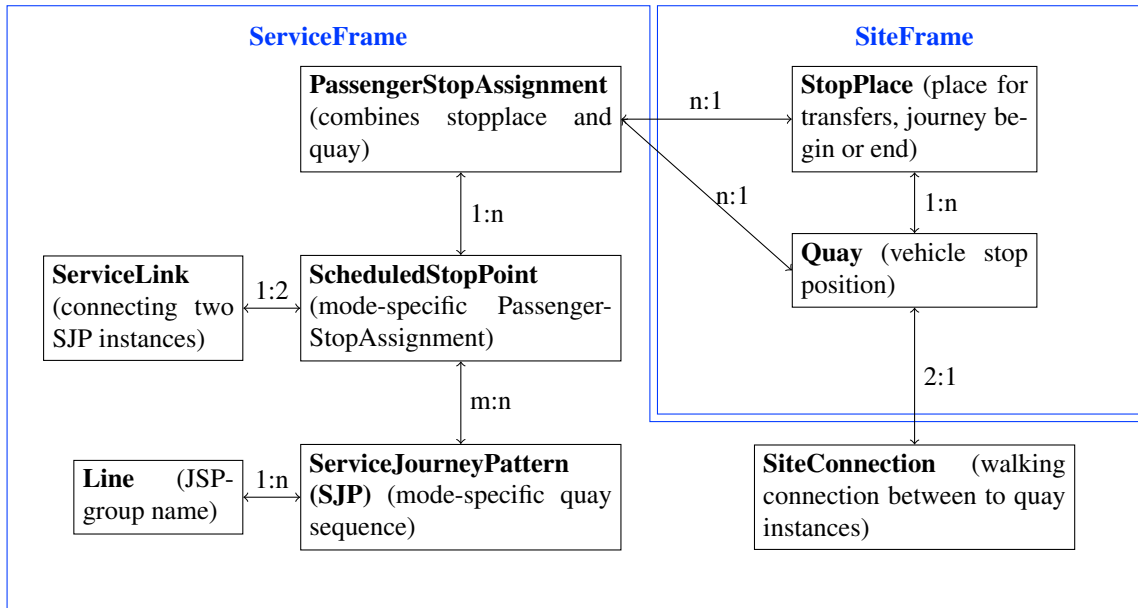


Fig. 1. Simplified ERM-model of the graph-defining NeTeX-data-objects

three values determine a specific stopplace. The fourth component is available to group all quays associated with a stop place; for example, it can specify a level (it is unused in this example since all quays are on the same level). The fifth component specifies a vehicle stop position (quay) at this stopplace. IFOPT-numbers enable a unique numbering scheme across all stopplaces and quays in a public transport network. These numbers are specified and used in the NeTeX-files.

3. Extracting graph parameters from NeTeX-files

3.1. Outline of the relevant NeTeX-data-objects and their relations among each other

Up to eight frames structures a NeTeX file. Two of these frames contain the graph-defining information: the SiteFrame as well as the ServiceFrame.

3.1.1. Node-defining data-objects stored in the SiteFrame

The set up of the node set of a graph requires the inspection of the data stored in the SiteFrame. The SiteFrame hosts two object collections. The first one contains a collection of **Quay**-instances. Each quay-instance¹ specifies a position where passengers can leave or enter a vehicle (vehicle stop post). The second collection maintains a list of **StopPlace**-instances. Every stopplace-instance is formed by a group of quay instances and represents a transfer location if at least two quay instances are associated with a stopplace-instance. Fig. 1 provides a simplified Entity-Relationship-Model (ERM) for the logical connections between the core objects stored in a NeTeX-file. The arc labels indicate the cardinalities of associated instances of the two involved objects.

3.1.2. Arc-defining data-objects stored in the ServiceFrame

The ServiceFrame includes the information needed to set up the arc set of a graph. A **PassengerStopAssignment**-instance (PSA-instance) merges a stopplace-instance with a quay-instance. The combination of a PSA-instance with

¹ some capitals are used in the object names in the NeTeX-standard. To ease the reading, we will use only the object names without capitals in subsequent appearances of the object names.

a specific transport mode defines a **ScheduledStopPoint**-instance (SSP-instance). It is used to declare a **ServiceJourneyPattern**-instance (SJP-instance). This object determines a sequence of consecutively visited quay-instances in the given transport mode (equivalent to a route). A **Line**-instance groups several SJP-instances and assigns a name to the group. It also instantiates a customer-usable transport service. A **SiteConnection**-instance describes the walking connection between two quay-instances to be used for transfers between an inbound and an outbound SJP-instance at a commonly served stopplace-instance. To store properties of the travel connection between two consecutively visited SSP-instances in a SJP-instance, a **ServiceLink**-object is used.

We can distinguish two types of graphs for passenger journey planning. A **structural graph** represents a condensed network, but the **routing graph** includes more detailed network information. To set up a structural graph, it is necessary to extract all stopplace instances stored in the selected NeTEx-files as nodes of the graph as well as the stored servicelink instances as arcs of the graph. For the setup of the routing graph the node set specification requires also the extraction of the quay-associated information as well as additionally fetching siteconnection instance information to compile the arc set.

3.2. Structural graph

If a potential passenger wants to know which lines to use and where to transfer between lines to move from the desired origin to the desired destination, connections between stopplaces are crucial. All stopplaces form the graph's node set. An arc $(i; j)$ originating from stopplace i and terminating in stopplace j is added to this graph if at least one service (route) travels from i to j nonstop. Such a graph is called a **structural graph**. Schematic line maps distributed to inform passengers how to travel through a network quickly are well-established examples of structural graphs.

3.2.1. Node-related information in the SiteFrame

The NeTEx-file collection contains the specifications for every relevant stopplace. For the compilation of the node set of the structural graph, we extract the location of a stopplace, which is the centroid of the geo-coordinates of all assigned quay instances, but also the stopplace name and its IFOPT number.

3.2.2. Arc-related information in the ServiceFrame

For the structural graph, we need information about those pairs of stopplaces where a non-stop service exists. This information is not coded directly into the NeTEx-files, but we can derive it by inspecting the service patterns describing the service routes. A sequence of consecutively visited quays defines a service pattern. If there is a service visiting first the quay with the IFOPT-number $A_1 : B_1 : C_1 : D_1 : E_1$ and then the next quay with the IFOPT-number $A_2 : B_2 : C_2 : D_2 : E_2$ we know that there is a non-stop service originating from the stopplace S_1 with IFOPT-number $A_1 : B_1 : C_1$ and terminating in the stopplace S_2 with the IFOPT-number $A_2 : B_2 : C_2$. We add the arc $(S_1; S_2)$ to the arc set of the structural graph.

3.3. Routing graph

The second type of journey planning graph is called **routing graph**. It also includes detailed information about the transfer activities of a passenger to enable a complete and detailed description of the travel path of a passenger through the network. The explicit modeling of transfers provides vital information, e.g., if a person with reduced mobility wants to avoid stairs during transfers but requires the existence of lifts instead. We must store the associated stopplace for every quay node as a node property. A **travel arc** connects two quay-nodes a and b if at least one service line offers nonstop transportation from a to b . We establish a **transfer arc** between each pair of two different quay nodes associated with the same stoplocation. The travel arcs and the transfer arcs form the arc set of a routing graph.

3.3.1. Node-related information in the SiteFrame

Besides the nodes representing the stopplaces, the routing graph specification requires adding all quays with their geo-coordinates, the name, and the IFOPT-code to the graph's node set. The hierarchy and association of quay instances to stop place instances is coded by the nested XML tags but can also be reconstructed from the used IFOPT-numbers.

3.3.2. Arc-related information in the ServiceFrame

We have to specify riding and transfer arcs. The specification of riding arcs follows the same idea as outlined for the structural graph. In case the inspection of the servicepatterns reveals that a service line connects the quay Q_1 with the IFOPT-number $A_1 : B_1 : C_1 : D_1 : E_1$ and the quay Q_2 with the IFOPT-number $A_2 : B_2 : C_2 : D_2 : E_2$ nonstop we add an arc to the arc set that originates from the node representing Q_1 and terminating in the node that represents Q_2 . Transfer connections are directly specified as SiteConnection instances in a NeTeX-file. We create a transfer arc for every found SiteConnection instance.

3.3.3. Routes

The identified servicepatterns enable the travel arc sequences specification. This process is the same as the one explained for the structural graph.

3.4. Limitations

We can use the derived graphs to determine passenger travel paths. However, NeTeX-files do not necessarily contain information about vehicle rotations. Therefore, we cannot use the derived graphs to route vehicles through the graph. In particular, if a car (or tram) finishes a duty at a quay Q^{end} and its next duty starts at a different quay Q^{start} , we do not have any information on how a vehicle moves from Q^{end} to Q^{start} .

We do not extract the timetable information stored in the NeTeX-files. For this reason, we cannot use the derived graph to determine a scheduled journey between two graph nodes (see also the outlook statement at the end of this manuscript).

4. Example application

We have developed a command line tool called `netex_xml_extractor` to extract those information relevant for defining a structural as well as a routing graph. Its sourcecode is available from the repository https://gitlab.hrz.tu-chemnitz.de/vbl/netex_xml_extractor.git (license CC-BY-NC).

To start the extraction it is necessary to specify the following parameters. First, a name for the network needs specification (SCENARIO). The definition of a collection of NeTeX-files to be inspected is the second required prerequisite to be executed before we can start the extraction tool.

4.1. Derived graphs and file storage

After the completion of the NeTeX-file inspection, the converter tool writes the collected graph information in seven files.

- `SCENARIO_stops.xml`: list of all identified stopplace instances indexed by IFOPT-numbers
- `SCENARIO_quays.xml`: list of all identified quay instances indexed by IFOPT-numbers
- `SCENARIO_sp.xml`: list of all identified servicepattern instances.
- `SCENARIO_structural_graph.xml`: XML-structured structural graph description
- `SCENARIO_structural_graph.tex`: LaTeX-source-code of a structural graph TikZ-picture
- `SCENARIO_routing_graph.xml`: XML-structured routing graph description
- `SCENARIO_routing_graph.tex`: LaTeX-source-code of a routing graph TikZ-picture
- `SCENARIO_routing_graph_bb.xml`: XML-structured routing graph description comprising only data contained in a user-specified bounding box
- `SCENARIO_routing_graph_bb.tex`: LaTeX-source-code of a routing graph TikZ-picture comprising only data contained in a user-specified bounding box

We have applied the tool to define graphs of the combined tram and bus network of Bremen, Germany (approx. 550.000 inhabitants). Overall, we can extract 23 NeTeX-files for tram services and 60 NeTeX-files for bus services from <https://www.opendata-oePNV.de> serving as Germany's National Access Point (publication date:

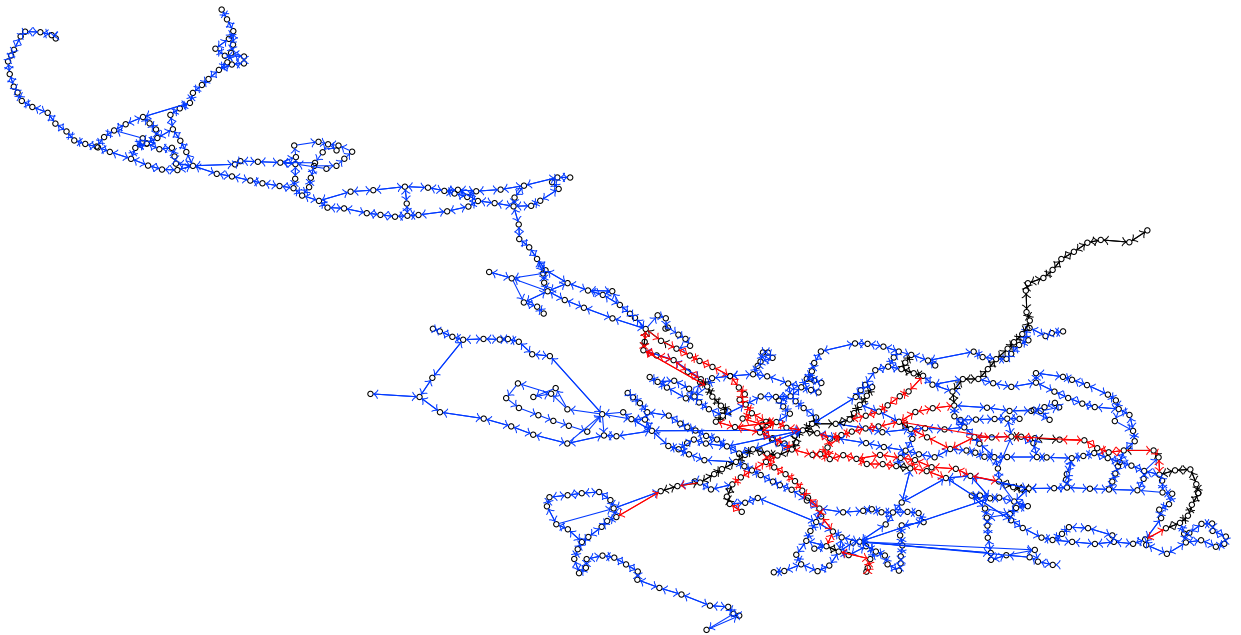


Fig. 2. Structural graph of the bus- & tram network in Bremen extracted from NeTeX-files.

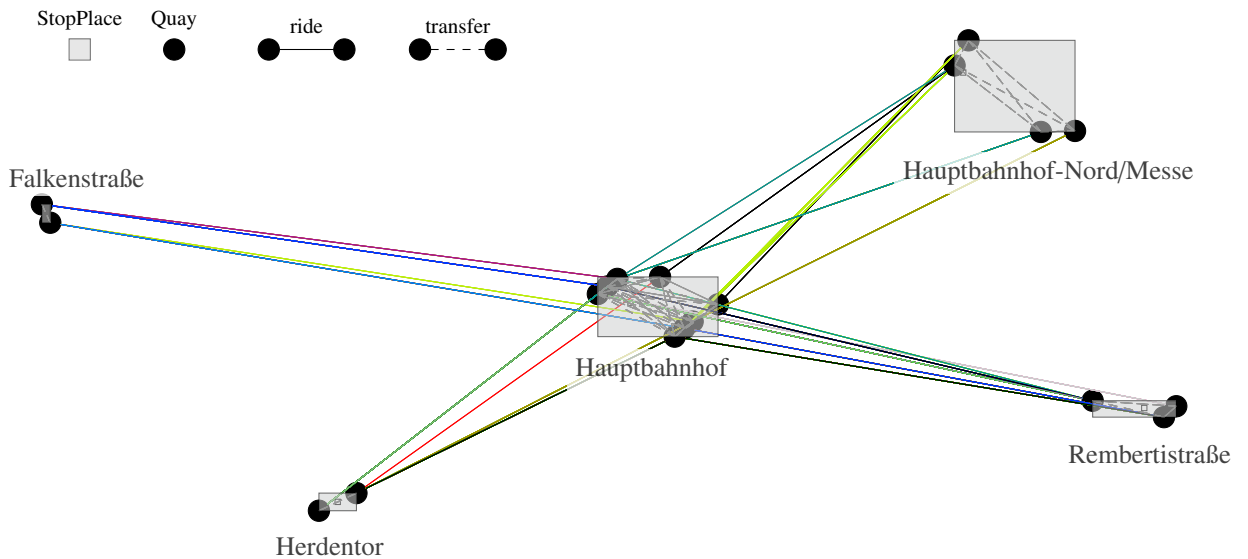


Fig. 3. Routing graph of the bus- & tram-network in Bremen: central stopplace instance Hauptbahnhof and adjacent stopplace instances in the bounding box $[8.80589; 8.82073] \times [53.07892; 53.0862178]$

30.12.2024). The operator of all lines is the company Bremer Straßenbahn AG (BSAG). They cover regular services, special services for events, pupil transport, and diverted services. Overall, we can extract 630 stopplaces that host 1803 quays. The converter identifies 954 servicepatterns for the analyzed bus- and tram routes.

The converter identifies 630 nodes and 1451 travel arcs to define the structural graph (Fig. 2). Arcs exclusively served by busses are colored in blue while arcs exclusively served by trams are shown in red. Arcs served by both types of vehicles are printed in black.

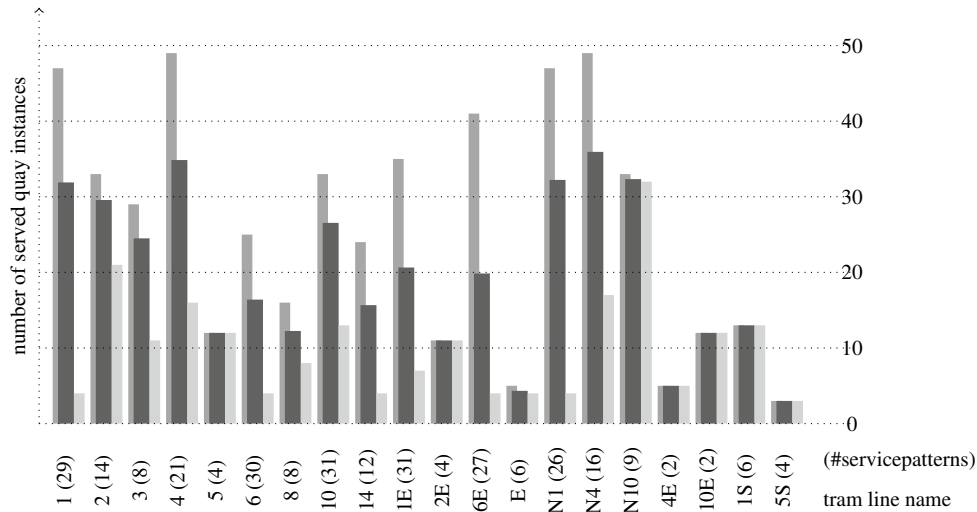


Fig. 4. Analysis of the identified service patterns (tram services only): maximal (gray bars), average (dark gray bars), and minimal number (light gray bars) of served quays (numbers in brackets indicate the associated service patterns for a service).

The full routing graph for the Bremen public transport network comprises 1813 nodes (representing the quays), 1743 travel arcs, 2608 walking arcs and 4 arcs declared as travel as well as walking arc. Fig. 3 shows a section of the resulting routing graphic around the central stopplace Hauptbahnhof (central rail station) considering a user-specified bounding box.

The gray-shaded areas represent stopplaces, and the small black circles are the quays associated with the stopplaces. Continuous arcs represent travel arcs. The color of a travel arc indicates the service line. The dashed arcs inside a stopplace area are the walking arcs.

4.2. Analysis of the service pattern collection

To support the analysis of the service patterns served by the service lines, the converter tool groups the identified service patterns by public service codes (line numbers published to the passengers). For every service code, it counts the number of associated service patterns. Besides, the converter calculates the minimum, the maximum, and the average number of quays served by the service patterns of a service line. The converter creates a file `SCENARIO_MODE_sp_quays.tex` for every identified transportation mode. It contains a `tikzpicture`-environment that visualizes the collected information.

Fig. 4 presents the service patterns analysis results of the tram line services in Bremen. A collection of 20 service lines has been extracted from the inspected 23 NeTEx-files. The number of associated service patterns varies between 4 and 31 among the lines. The dark bar height over each service line represents the average served quay instances in such a service pattern. It falls into the interval between 3 and ≈ 43 quays. The height of the mid-gray bar shows the maximal number of served quays in a service pattern of its service line (between 3 and 49 quays). Similarly, the light gray bars indicate the shortest service patterns per service line. Here, we can observe variations from 3 to 32 quays.

5. Summary and outlook

We have explained the parameterization of two variants of a mathematical graph to represent a real-world multi-modal public transport network exploiting only information found in NeTEx-files. The developed converter software automatically creates different output files containing node sets, arc sets, and routes.

Inconsistencies in the processed data are detected automatically if a quay instance cannot be assigned to a stopplace or if a stopplace does not possess a single quay instance. However, the validation with respect to completeness and

accuracy of the extracted data is still an open issue. One of the next research steps will address the comparison of the NeTEx-data with the data found in an equivalent GTFS-feed.

Another issue that requires further research is the visualization of lines serving an arc. Currently, it happens that those arcs are overlapping and cannot be seen (Fig. 3). There are algorithms to automatically identify auxiliary origin and destination node coordinates to arrange the overlapping arcs side by side. For an example, we refer to Brosi and Bast (2023).

The third planned research direction addresses the integration of the timetable information stored in the `TimetableFrame` of a NeTEx-file. Its useful integration into the created output files will support researchers working in the field of algorithms for passenger journey planning.

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